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IPM Fact Sheet Series

***UMass Extension Fruit Team***

*Fact Sheet #AD-003*

# **Apple - Cedar Apple Rust** (*Gymnosporangium juniperi-virginianae*)

## Overview

Cedar-apple rust is a fungal disease that requires both a Rosaceaous host (such as apple) and a cedar host (such as Eastern red cedar). On apples, bright orange-yellow lesions are first visible after bloom. These lesions develop from spores released from galls on cedars earlier in the spring. Some species of cedar-apple rusts also produce lesions on fruit.

## Symptoms

The most conspicuous symptoms of Cedar Apple Rust (CAR) on apple are bright orange lesions on the leaves. Lesions may have red or yellow borders. Cedar-apple rust appears on fruit first as bright orange, slightly raised lesions that become brown and cracked as the fruit enlarges. Stem infection causes a slight swelling of the stem and may result in abscission of the young fruit. On eastern red cedar (*Juniperus virginianae*) cedar-apple rust produces brown, globular galls ranging in size from ¼" to 2 inches in diameter. These are dimpled like a golf ball in the dormant season, but produce gelatinous, orange fruiting bodies (called telia or telial horns) during spring rains.

Infections appear on apple as bright yellow-orange lesions on the upper surface of leaves, petioles, and young fruit. Lesions that occur on fruit are superficial and do not lead to fruit rot but do reduce marketability of the fruit.

## Disease Cycle

Telia release spores that can infect apple trees during wet weather. These spores may be carried by the wind for more than 1 km (0.6 mile) to vulnerable apple tissue. The youngest leaves are the most vulnerable to infection: susceptibility declines as tissues mature. One to two weeks after infection, lesions appear on the upper sides of apple leaves. One to two months later, lesions appear on the undersides of the leaves. These lesions bear small, tubular protrusions (aecia) that produce spores which are dispersed by wind. These spores are incapable of infecting apples, but those that land on eastern red cedar can infect young leaves and initiate gall formation. Unlike the galls of some other rust fungi, those of *G. juniperi-virginianae* are capable of producing telia only once. The fungus requires both the eastern red cedar and apple in order to complete its life cycle.



**Figure 1)** Left, CAR gall on cedar in early May; middle, CAR lesions on apple leaves in late May; right, mature CAR lesion on apple leaves in July. Photos: H. Faubert URI. [**Source**: New England Tree Fruit Management Guide at: netreefruit.org]

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**Figure 2)** Cedar apple rust lesion on fruit. **Photo**: D. Rosenberger, Cornell Univ. [**Source**: New England Tree Fruit Management Guide at: netreefruit.org]

## Management Strategies

**Planting Resistant Cultivars:** Apple cultivars with **resistance** to cedar-apple rust include: Baldwin, Delicious (red), Empire, Enterprise, Gala Supreme, Jerseymac, Keepsake, Liberty, McIntosh, Milton, Niagara, Paulared, Redfree, Regent, Sansa, Spartan, Sundance, Viking, and Zestar!.

Apple cultivars that are **very susceptible** to cedar-apple rust include: Ambrosia, Braeburn, Cameo, Chinook, Crimson Crisp, Fuji, Gala, Ginger Gold, Golden Delicious, Jonathan, Lodi, Prima, Rome Beauty, Shizuka, Spigold, Twenty Ounce, Wealthy, Winter Banana, and York Imperial.

***Monitoring***: Scout vicinity for symptomatic cedars. Spore production and release from galls on cedars are favored by wet weather during May and June. Scout orchard for infections in mid-summer.

## Control Strategies

*Cultural/Biological:*

- Reduce inoculum sources by cutting down nearby eastern red cedar.
- Plant resistant cultivars when possible.
- Prune trees to open the canopy to light, air, and spray penetration.

## Chemical

- Refer to the New England Tree Fruit Management Guide for specific materials and rates recommended for managing Cedar Apple Rust.
- Apply recommended protectant fungicides during bloom if symptoms were found in the previous year, especially if current weather is warm and wet.
- Make follow-up applications as indicated by scouting and weather conditions until mid-season when cedar galls cease producing spores.
- Rotate fungicide materials from different FRAC groups to avoid promoting the development of resistant strains of this disease.

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**Additional information available on the MYIPM app:** https://apps.bugwood.org/apps/myipmseries/

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